

Figure 3.1 The sensor signal chain by which a change in the physical environment makes its way to the digital twin representation via an electronic sensor.

1. *Physical change*—something in the environment changes (temperature rises, pressure increases, object moves).
2. *Transduction*—the sensor's sensing element converts this physical change into an electrical signal, typically a voltage change.
3. *Signal conditioning*—the raw signal gets amplified and filtered to remove noise.
4. *Analog-to-digital conversion*—the voltage is sampled and converted to digital values.
5. *Processing*—digital signal processing performs offset and gain correction, linearization and digital filtering.
6. *Communication*—formatted data is transmitted to your digital twin platform.

Consider measuring temperature, which is something I will do throughout my home. A thermistor in a temperature sensor I have installed varies its resistance based on the ambient temperature, producing a tiny proportional voltage in a circuit. This millivolt signal needs amplification to a usable level. An analog-to-digital converter (ADC) then samples this voltage, converting each sample to a digital number. Firmware in the sensor applies calibration curves to convert raw ADC counts to actual temperature values, then formats this data for transmission.